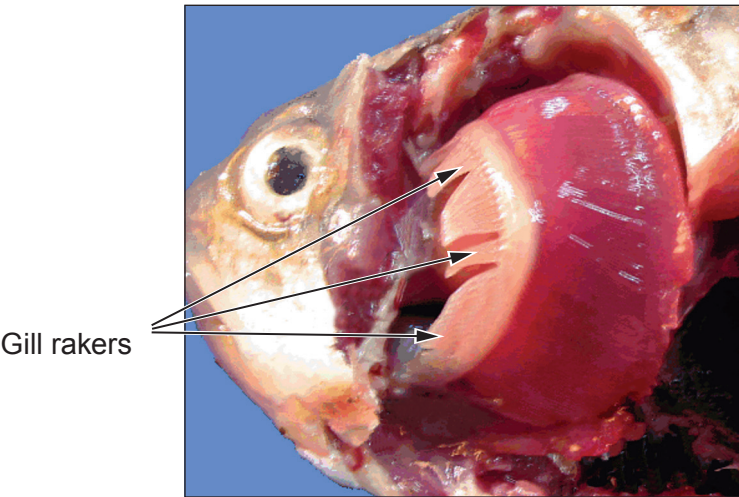
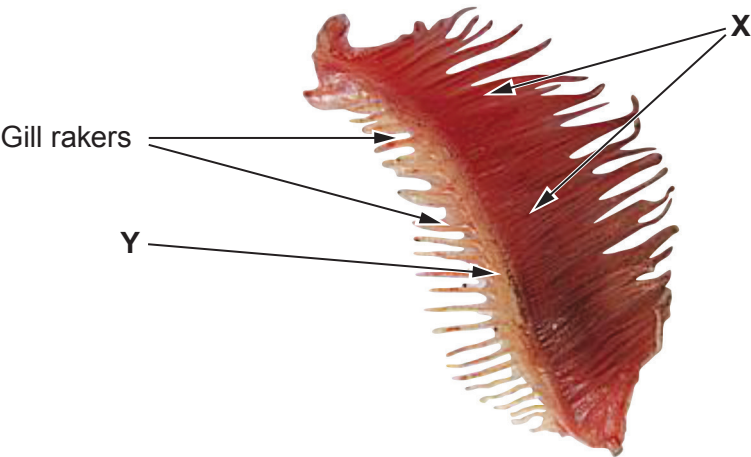


4. **Photograph A** shows a freshly dissected bony fish with the operculum removed to show the gills.
Photograph B shows a single gill 30 minutes after having been removed from the fish.

Photograph A



Photograph B



- (a) (i) Identify the structures labelled **X** and **Y** in **Photograph B**. [1]
- X**
- Y**
- (ii) Suggest the function of the gill rakers. [1]
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(iii) Use the photographs, and your knowledge, to explain why fish suffocate when out of water. [4]

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(b) Gas exchange in bony fish uses the countercurrent flow mechanism, where blood flows through the capillaries of the gill lamellae in the opposite direction to water flowing across them. In parallel flow, blood flows through the capillaries of the gill lamellae in the same direction as water flowing across them. Explain the advantages of the countercurrent flow mechanism compared to the parallel flow mechanism. [4]

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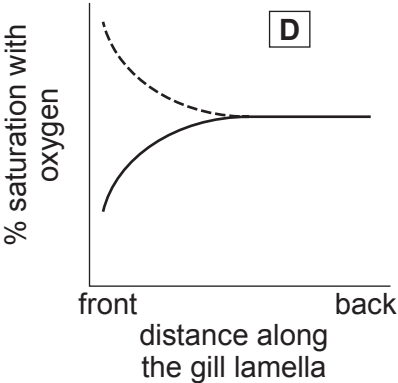
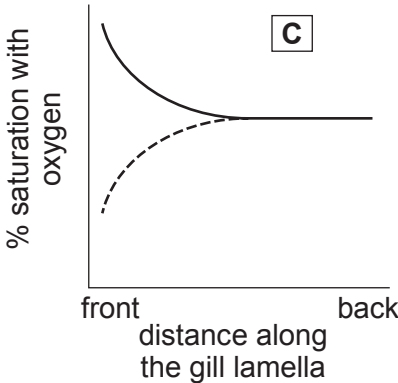
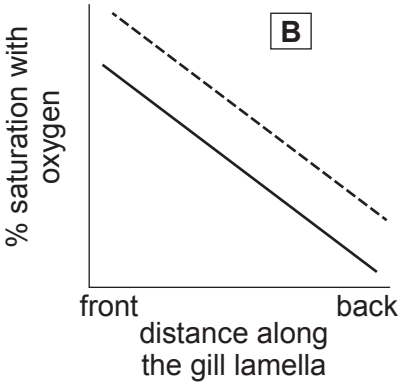
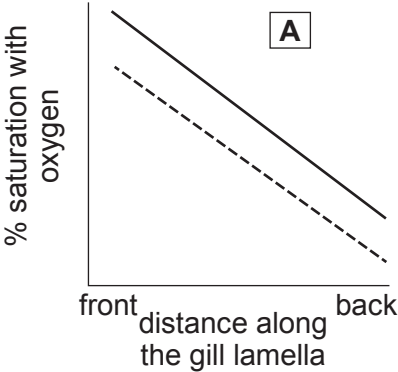
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(c) The graphs below show representations of changes in oxygen concentration as water flows across the gill lamellae.



Key: — = water; - - - - - = blood

(i) Identify which graph illustrates parallel flow and which graph illustrates countercurrent flow. [1]

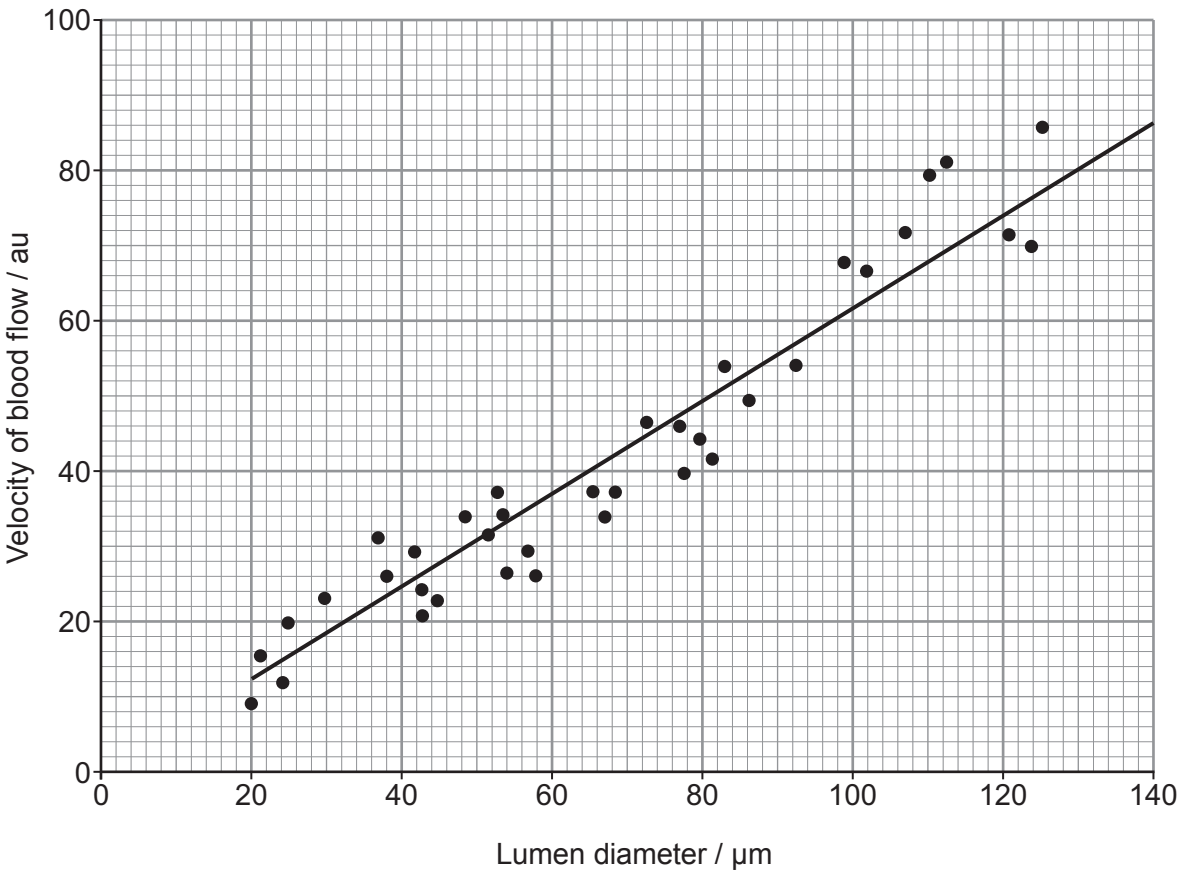
Parallel flow Countercurrent flow

(ii) **Clearly** insert arrows on the graph which represents countercurrent flow to show the direction of blood flow **and** of water flow. [1]



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(d) The scatter diagram below shows the relationship between blood vessel lumen diameter and velocity of blood flow.



(i) State the type of correlation between the two variables shown on the scatter diagram. [1]

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(ii) Use the graph to determine the gradient of the line of best fit. [2]

gradient =

